CIVICPULSE API GUIDE

Endpoints first. Plain-language bullet explanations.

AUTHENTICATION: HOW THE USERS WORK

There are four user role types in the code. This is not the number of registered accounts. The actual account count depends on the current database; it cannot be determined from the source code or this guide.

- resident - the default role for anyone who registers. Can submit, browse and follow reports, and use personal notifications.
- verifier - reviews reports, verifies or rejects them, assigns verified cases, reviews duplicate suggestions, and can list authority users.
- authority - works on assigned cases, moves them to in_progress and resolved, views staff analytics, and can list authority users.
- admin - staff role with broad access, including user listing, duplicate decisions, analytics and hotspot recomputation. Admin accounts are provisioned separately.

SIGN-IN FLOW

- Registering creates a resident account only; a submitted role is not used. Email is normalized, passwords are stored as hashes, and duplicate emails are rejected.
- Login checks the email and password, then returns a signed JWT token and user profile. The token lasts 7 days.
- For protected endpoints, send Authorization: Bearer <token>. The backend validates the token and reloads the user and role from the database on each request.
- 401 means the request is unauthenticated (missing/invalid/expired token). 403 means signed in, but the role is not allowed.
- Roles are not self-selected at signup. Staff users can be created with the Flask CLI command ‘flask create-user‘ and the required role.
- There is no live API route to change a user’s role. The OpenAPI document lists PATCH /users/{id}/role, but the Flask app does not register it.

POST /api/v1/auth/register | Public

- Input: name, email, password (at least 8 characters), optional area_id. Success: 201 with {token, user}; invalid data: 422; existing email: 409.

POST /api/v1/auth/login | Public

- Input: email and password. Success: 200 with {token, user}; incorrect email or password returns 401 with the same message.

GET /api/v1/auth/me | Signed-in user

- Returns the current user’s profile from the authenticated token and database.

GET /api/v1/users | Verifier, authority, admin

- Admins can list users and filter by role. Verifiers and authorities only receive authority users, with email removed. Results are paginated.
- This route does not return an account total. To know how many accounts exist, count records in the users table or add a dedicated count/reporting route.

REFERENCE DATA

GET /api/v1/categories | Public

- Returns available report categories and icons. The IDs are used by report submission.

GET /api/v1/areas | Public

- Returns area IDs and names. A report is assigned an area only if its location falls inside a configured area boundary.

REPORT CREATION AND DISCOVERY

POST /api/v1/reports | Signed-in user

- Send multipart form fields: category_id, lat, lon, photo (JPEG or PNG required), and optional description (maximum 500 characters).
- The API validates the category, coordinates and actual image signature; stores the photo; saves the report as reported; and records its first status event.
- After saving, it queues optional AI processing. The resident gets the saved report without waiting for AI. Queue failure does not undo the report.

GET /api/v1/reports | Signed-in user

- Returns a paginated list, newest first. Filters: bbox, category_id, area_id, status, mine=true, following=true, include_duplicates=true.
- Merged duplicates are hidden by default and represented by their parent case. Page size defaults to 50 and is capped at 200.

GET /api/v1/reports/{id} | Signed-in user

- Returns one report: category, status, coordinates, photo URL, creator/assignee IDs, due date, duplicate count/link, AI suggestion if present, and current user’s following flag.

POST /api/v1/reports/{id}/follow | Signed-in user

- Follows the report so the user can receive its updates. Safe to repeat; returns 204.

DELETE /api/v1/reports/{id}/follow | Signed-in user

- Stops following the report. Safe when already unfollowed; returns 204.

GET /uploads/{path} | Public; outside /api/v1

- Serves a saved report photo. Report responses provide the photo_url used to request it.
- Coordinates in API responses use {lat, lon}; PostGIS storage writes points as longitude then latitude.

REPORT WORKFLOW

Normal path: reported -> verified -> assigned -> in_progress -> resolved. A verifier or admin can reject a reported case instead. Rejected and resolved are final states.

PATCH /api/v1/reports/{id}/status | Verifier, authority, admin

- JSON input: {status, note?}. Verifier/admin can verify or reject; rejecting requires a reason.
- Only the assigned authority or an admin can move assigned -> in_progress -> resolved. Use the assign endpoint to enter assigned.
- Each accepted change updates the report and adds a status-history event; interested users are notified in the same transaction.
- Invalid transition returns 409. A merged duplicate cannot be changed directly; update its parent case.

POST /api/v1/reports/{id}/assign | Verifier, authority, admin

- JSON input: {assignee_id, due_date?}. The selected user must have the authority role; due_date must be YYYY-MM-DD and not in the past.
- Assigning a verified report moves it to assigned and records the assignee/due date. An assigned but not started report can be reassigned; in-progress work cannot.
- Assignment is part of the workflow: it records an event and creates notifications for the new assignee and, when reassigned, the former assignee.

GET /api/v1/reports/{id}/history | Signed-in user

- Returns status events in time order, including who changed the status and any note. Used as the case audit trail and to calculate resolution time.

WHO CAN CHANGE STATUS?

- reported -> verified or rejected: verifier/admin.
- verified -> assigned: verifier/authority/admin, through POST /assign.
- assigned -> in_progress -> resolved: assigned authority or admin.

DUPLICATE REVIEW

AI produces suggestions; a human verifier/admin decides whether reports are actually the same case. A merge does not delete the child report.

GET /api/v1/reports/{id}/duplicates | Verifier, admin

- Returns saved candidate matches, highest score first, with reasons. Suggestions only exist after the duplicate AI hook has run successfully.

POST /api/v1/reports/{id}/merge | Verifier, admin

- JSON input: {parent_id}. The path ID is the duplicate; parent_id is the main report to keep as the case.
- Only reported/verified children can be merged, and the parent must still be open. Followers and the child's author follow the parent.
- Updates duplicate links and parent report_count, writes an audit note, and alerts the child author. Default report lists then omit the child.

POST /api/v1/reports/{id}/unmerge | Verifier, admin

- Restores a merged report as its own case, clears the duplicate link, adjusts the parent's count and records an audit note.

BACKGROUND AI AND JOB QUEUE

- After report creation: API saves the report -> enqueues process_new_report -> worker can classify the photo and find nearby duplicate candidates -> results are saved for the report/staff review.
- AI hooks are optional configuration. If not configured, classification and duplicate suggestions are skipped; an AI category is not guaranteed.
- Duplicate matching considers eligible older, open, unmerged nearby reports. Hotspot refresh may also run in the background.
- GET /api/v1/health exposes queue status. A growing pending count with no workers points to a worker/Redis issue, not a failed report save.

NOTIFICATIONS

The backend creates messages when relevant report actions happen. A user is not notified about their own action. Notifications are scoped to the signed-in user.

GET /api/v1/notifications | Signed-in user

- Returns the current user's notifications, newest first. Optional filters: unread=true, page, per_page.

GET /api/v1/notifications/unread-count | Signed-in user

- Returns {count}; intended for the app's unread badge.

POST /api/v1/notifications/{id}/read | Signed-in user

- Marks one of the current user's notifications read. A notification owned by someone else is treated as not found.

POST /api/v1/notifications/read-all | Signed-in user

- Marks all of the current user's notifications read.
- Status changes notify the report author/followers; assignment notifies the assignee; reassignment also notifies the former assignee; merge notifies the child author.

STAFF ANALYTICS

All analytics routes below require authority or admin. Counts generally represent cases, so merged duplicate children are not counted twice. Trends count every submission, including duplicates.

GET /api/v1/analytics/summary | Authority, admin

- Dashboard totals, open/resolved counts, hotspot count, category/status breakdowns, open cases by area, and average days to resolve. Optional area_id.

GET /api/v1/analytics/categories | Authority, admin

- Case count for each category, including zero counts. Optional area_id.

GET /api/v1/analytics/statuses | Authority, admin

- Case count by status for statuses that currently have cases. Optional area_id.

ANALYTICS, HOTSPOTS AND SERVICE STATUS

GET /api/v1/analytics/trends | Authority, admin

- Daily report submissions in UTC, with zero-count days included. days defaults to 30 and must be between 1 and 365. Duplicate submissions are included.

GET /api/v1/analytics/hotspots | Authority, admin

- Returns saved geographic clusters, largest first: center, radius, report count, most common category and computation time.

POST /api/v1/analytics/hotspots/recompute | Admin

- Rebuilds clusters on demand. Optional JSON: {eps_m, min_samples}; defaults are 250 metres and 3 reports. Background report processing may refresh them automatically too.

GET /api/v1/health | Public

- Checks API/database/PostGIS and reports queue mode/health where available. Useful for diagnosing backend or worker status.

HOW THE PIECES CONNECT

- Auth identifies the caller and role. Reports use that identity as creator and gate staff actions by role.
- Report status changes write status_events and trigger notifications; the history endpoint reads that audit trail.
- Follow records decide who receives report updates. Merging moves the child audience to the parent case.
- The worker enriches reports asynchronously with optional AI outputs and may refresh hotspots. Staff analytics read persisted reports/events/hotspots.

API CONVENTIONS AND ACCURACY NOTES

- Base path is /api/v1 except photo delivery at /uploads/{path}. Protected routes require a bearer token.
- Errors follow {error: {code, message, details}}. Typical meanings: 401 unauthenticated, 403 forbidden, 404 missing, 409 conflicting state, 422 invalid input.
- Four role types are defined; the number of actual user accounts is database data. GET /users returns a page, not a total count.
- PATCH /api/v1/users/{id}/role is documented in OpenAPI but is not registered in the running Flask app. Do not present it as available.